上海科技大学
ShanghaiTech University

# CS240 Algorithm Design and Analysis

## Lecture 17

## Local Search

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# Local Search

Gradient descent

Maximum Cut

Nash Equilibria

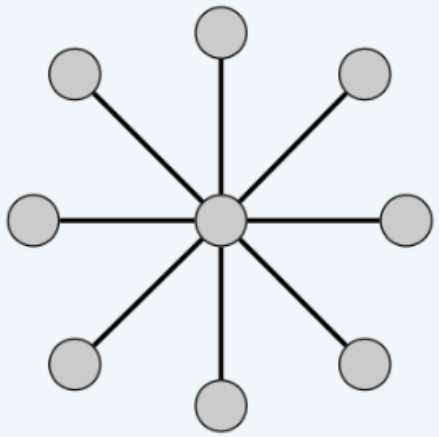
# Local Search: Gradient Descent

# Gradient Descent: Vertex Cover

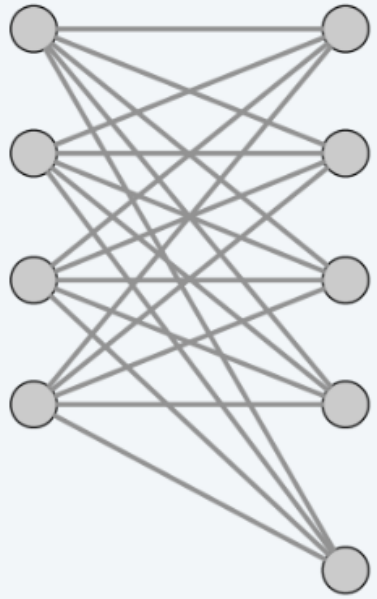
- **Vertex cover.** Given a graph G = (V, E), find a subset of nodes S of minimal cardinality such that for each (u, v) ∈ E, either u or v (or both) are in S
- **Neighbor relation.** S ∼ S' if S' can be obtained from S by adding or deleting a single node. Each vertex cover S has at most n neighbors
- **Gradient descent.** Start with S = V. If there is a neighbor S' that is a vertex cover and has lower cardinality, replace S with S'
- **Remark.** Algorithm terminates after at most n steps since each update decreases the size of the cover by one

# Gradient Descent: Vertex Cover

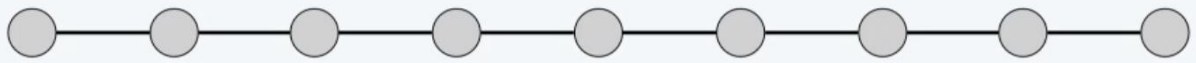
- **Local optimum.** No neighbor is strictly better



Optimum = center node only
Local optimum = all other nodes



Optimum = all nodes on left side
Local optimum = all nodes on right side



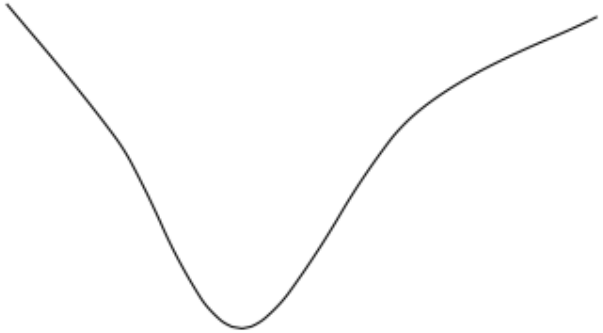
optimum = even nodes
local optimum = omit every third node

# Local Search

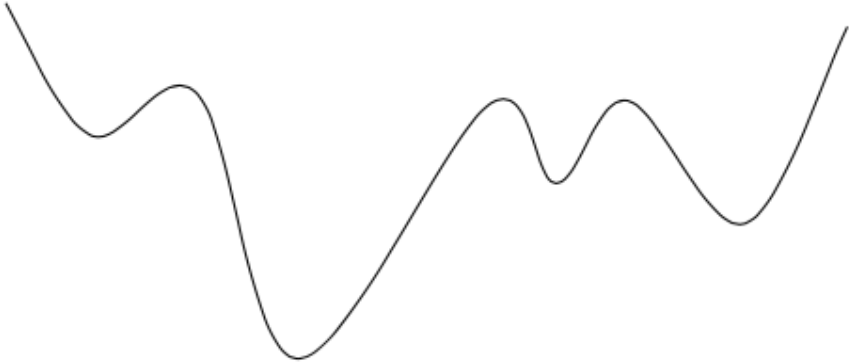
**Local search.** Algorithm that explores the space of possible solutions in sequential fashion, moving from a current solution to a "nearby" one.

**Neighbor relation.** Let $S \sim S'$ be a neighbor relation for the problem.

**Gradient descent.** Let S denote current solution. If there is a neighbor S' of S with strictly lower cost, replace S with the neighbor whose cost is as small as possible. Otherwise, terminate the algorithm.



A funnel



A jagged funnel

# Local Search: Maximum Cut

# Maximum Cut

**Maximum cut.** Given an undirected graph G = (V, E) with positive integer edge weights $w_e$, find a node partition (A, B) such that the total weight of edges crossing the cut is maximized.

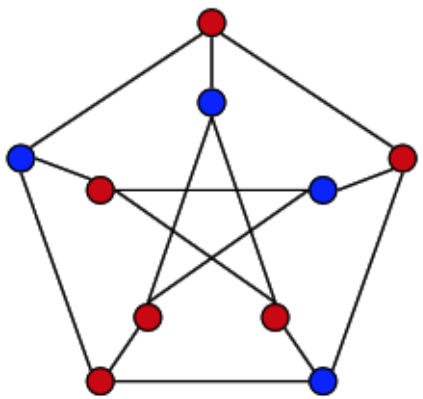
$$w(A,B) := \sum_{u \in A,\, v \in B} w_{uv}$$

**Note:**

- The Min-Cut problem can be solved in poly time.
- The Max-Cut problem is NP-hard.

**Toy application**

- n activities, m people
- Each person wants to participate in two of the activities
- Schedule each activity in the morning or afternoon to maximize number of people that can enjoy both activities



- **Real applications.** Circuit layout, statistical physics

# Maximum Cut

**Single-flip neighborhood.** Given a partition (A, B), move one node from A to B, or one from B to A if it improves the solution.

**Local search algorithm.**

```
Max-Cut-Local (G, w) {
   Pick a random node partition (A, B)

   while (∃ improving node v) {
      if (v is in A) move v to B
      else           move v to A
   }

   return (A, B)
}
```

# Maximum Cut: Local Search Analysis

**Theorem.** Let (A, B) be a locally optimal partition and let (A*, B*) be the optimal partition. Then $w(A, B) \geq \frac{1}{2} \Sigma_e w_e \geq \frac{1}{2} w(A^*, B^*)$.

Weights are nonnegative

**Pf.**

- Local optimality implies that for any $u \in A$ : $\sum_{v \in A} w_{uv} \leq \sum_{v \in B} w_{uv}$
  Adding up all these inequalities yields:

$$2 \sum_{\{u,v\} \subseteq A} w_{uv} \leq \sum_{u \in A,\ v \in B} w_{uv} = w(A,B)$$

- Similarly

$$2 \sum_{\{u,v\} \subseteq B} w_{uv} \leq \sum_{u \in A,\ v \in B} w_{uv} = w(A,B)$$

- Now,

each edge counted once

$$\sum_{e \in E} w_e = \underbrace{\sum_{\{u,v\} \subseteq A} w_{uv}}_{\leq \frac{1}{2} w(A,B)} + \underbrace{\sum_{u \in A, v \in B} w_{uv}}_{w(A,B)} + \underbrace{\sum_{\{u,v\} \subseteq B} w_{uv}}_{\leq \frac{1}{2} w(A,B)} \leq 2w(A,B)$$

∎

# Maximum Cut: Big Improvement Flips

**Local search.** Within a factor of 2 for MAX-CUT, but not poly-time!

**Big-improvement-flip algorithm.** Only choose a node which, when flipped, increases the cut value by at least $\frac{2\varepsilon}{n} w(A, B)$

**Claim.** Upon termination, big-improvement-flip algorithm returns a cut (A, B) with $(2 + \varepsilon)$ $w(A, B) \geq w(A^*, B^*)$.

**Pf idea.** Add $\frac{2\varepsilon}{n} w(A, B)$ to each inequality in original proof.

**Claim.** Big-improvement-flip algorithm terminates after $O(\varepsilon^{-1} n \log W)$ flips, where $W = \Sigma_e w_e$.

- Each flip improves cut value by at least a factor of $(1 + \varepsilon/n)$.
- After $n/\varepsilon$ iterations the cut value improves by a factor of 2
- Cut value can be doubled at most $\log_2 W$ times.

if $x >= 1$, $(1+1/x)^x >= 2$

# Nash Equilibria

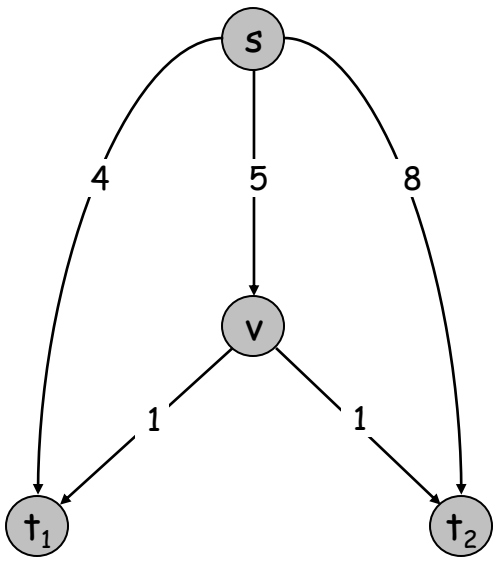
# Multicast Routing with Fair Cost Sharing

**Multicast routing.** Given a directed graph $G = (V, E)$ with edge costs $c_e \geq 0$, a source node $s$, and $k$ agents located at terminal nodes $t_1, \ldots, t_k$.

Agent $j$ must construct a path $P_j$ from node $s$ to its terminal $t_j$.

**Fair share.** If $x$ agents use edge $e$, they each pay $c_e / x$.

| 1 | 2 | 1 pays | 2 pays |
|---|---|---|---|
| outer | outer | 4 | 8 |
| outer | middle | 4 | 5 + 1 |
| middle | outer | 5 + 1 | 8 |
| middle | middle | 5/2 + 1 | 5/2 + 1 |



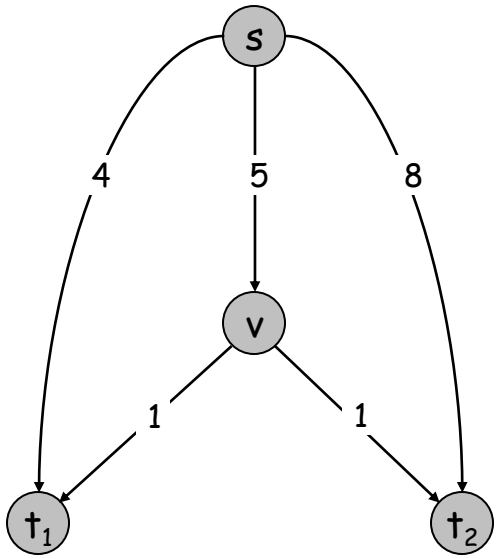
# Nash Equilibrium

**Best response dynamics.** Each agent is continually prepared to improve its solution in response to changes made by other agents.

**Nash equilibrium.** Solution where no agent has an incentive to switch.

**Ex:**

- Two agents start with outer paths.
- Agent 1 has no incentive to switch paths (since 4 < 5 + 1), but agent 2 does (since 8 > 5 + 1).
- Once this happens, agent 1 prefers middle path (since 4 > 5/2 + 1).
- Both agents using middle path is a Nash equilibrium.

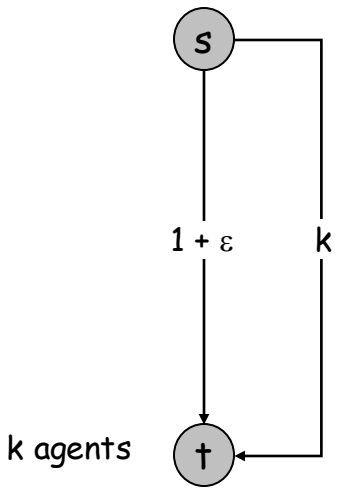



**Note.** Best response dynamics may not terminate since no single objective function is being optimized.

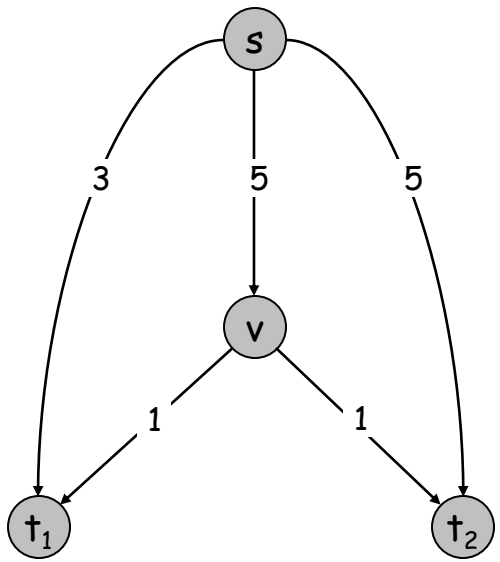
# Social Optimum

**Social optimum.** Minimizes total cost to all agents.

**Observation.** In general, there can be many Nash equilibria. Even when it's unique, it does not necessarily equal the social optimum.




Social optimum = 1 + ε
Nash equilibrium A = 1 + ε
Nash equilibrium B = k




Social optimum = 7
Unique Nash equilibrium = 8

# Price of Stability

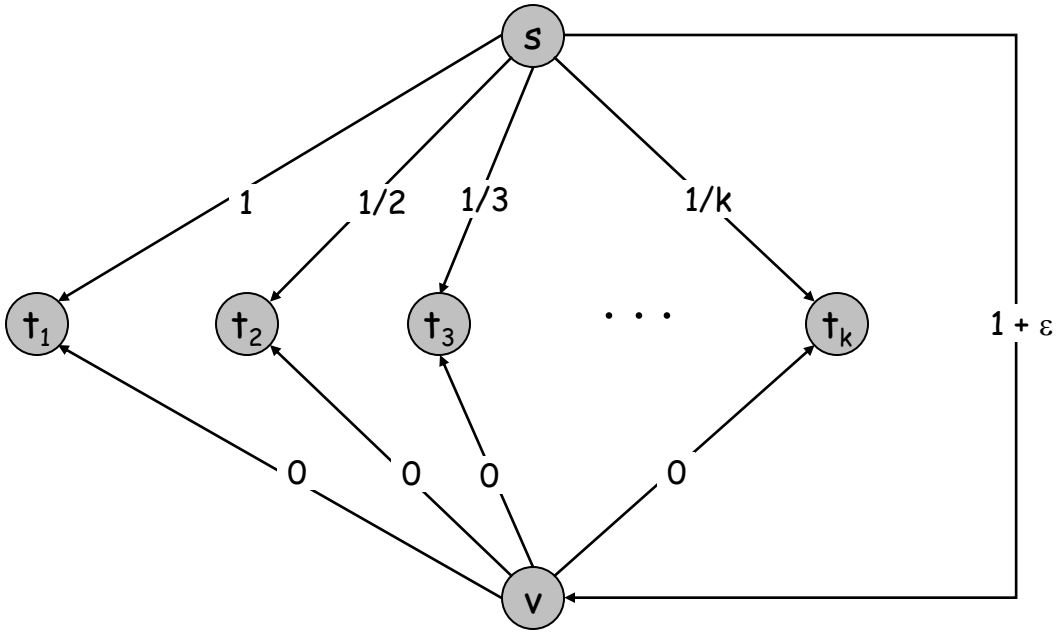
**Price of stability.** Ratio of best Nash equilibrium to social optimum.

**Fundamental question.** What is price of stability?

**Ex:** Price of stability = $\Theta(\log k)$.

- **Social optimum:** Everyone takes bottom paths.
- **Unique Nash equilibrium:** Everyone takes top paths.
- **Price of stability:** $H(k) / (1 + \varepsilon)$.

**$1 + 1/2 + \ldots + 1/k$**



# Finding a Nash Equilibrium

**Theorem.** The following algorithm terminates with a Nash equilibrium (but its running time may be exponential).

```
Best-Response-Dynamics(G, c) {
   Pick a path for each agent
   while (not a Nash equilibrium) {
      Pick an agent i who can improve by switching paths
      Switch path of agent i
   }
}
```

$H(0) = 0$

$H(k)=\sum_{i=1}^{k} \frac{1}{i}$

**Pf.** Consider a set of paths $P_1, \ldots, P_k$.

- Let $x_e$ denote the number of paths that use edge e.
- Let $\Phi(P_1, \ldots, P_k) = \Sigma_{e \in E}\, c_e \cdot H(x_e)$ be a potential function.
- Since there are only finitely many sets of paths, it suffices to show that $\Phi$ strictly decreases in each step.

# Finding a Nash Equilibrium

**Pf.** (continued)

- Consider agent j switching from path $P_j$ to path $P_j'$.
- Agent j switches because

$$\underbrace{\sum_{f \in P_j' - P_j} \frac{c_f}{x_f + 1}}_{\text{newly incurred cost}} < \underbrace{\sum_{e \in P_j - P_j'} \frac{c_e}{x_e}}_{\text{cost saved}}$$

- $\Phi$ increases by

$$\sum_{f \in P_j' - P_j} c_f \left[ H(x_f + 1) - H(x_f) \right] = \sum_{f \in P_j' - P_j} \frac{c_f}{x_f + 1}$$

- $\Phi$ decreases by

$$\sum_{e \in P_j - P_j'} c_e \left[ H(x_e) - H(x_e - 1) \right] = \sum_{e \in P_j - P_j'} \frac{c_e}{x_e}$$

- Thus, net change in $\Phi$ is negative. ▪

# Bounding the price of stability

**Claim.** Let $C(P_1, \ldots, P_k)$ denote the total cost of selecting paths $P_1, \ldots, P_k$.

For any set of paths $P_1, \ldots, P_k$, we have

$$C(P_1, \ldots, P_k) \;\le\; \Phi(P_1, \ldots, P_k) \;\le\; H(k) \cdot C(P_1, \ldots, P_k)$$

**Pf.** Let $x_e$ denote the number of paths containing edge e.

- Let $E^+$ denote set of edges that belong to at least one of the paths.

$$C(P_1, \ldots, P_k) \;=\; \sum_{e \in E^+} c_e \;\le\; \underbrace{\sum_{e \in E^+} c_e \, H(x_e)}_{\Phi(P_1, \ldots, P_k)} \;\le\; \sum_{e \in E^+} c_e \, H(k) \;=\; H(k) \; C(P_1, \ldots, P_k)$$

# Bounding the price of stability

**Theorem.** There is a Nash equilibrium for which the total cost to all agents exceeds that of the social optimum by at most a factor of H(k).

**Pf.**

- Let $(P_1^*, \ldots, P_k^*)$ denote set of socially optimal paths.
- Run best-response dynamics algorithm starting from P*.
- Since $\Phi$ is monotone decreasing $\Phi(P_1, \ldots, P_k) \leq \Phi(P_1^*, \ldots, P_k^*)$.

$$C(P_1, \ldots, P_k) \leq \Phi(P_1, \ldots, P_k) \leq \Phi(P_1^*, \ldots, P_k^*) \leq H(k) \cdot C(P_1^*, \ldots, P_k^*)$$

previous claim applied to P

previous claim applied to P*

# Summary

**Existence.** Nash equilibria always exist for k-agent multicast routing with fair sharing.

**Price of stability.** Best Nash equilibrium is never more than a factor of H(k) worse than the social optimum.

**Big open problem.** Find any Nash equilibrium in poly-time